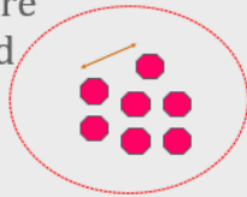


[S8] Cluster analysis I

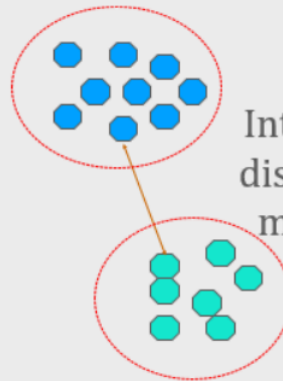
CLUSTER ANALYSIS: BASIC CONCEPTS

- **Cluster analysis** or **clustering** is the task of grouping a set of data objects in such a way that objects in the same group (called a cluster) are more similar (in some sense or another) to each other than to those in other groups (clusters).

Intra-cluster
distances are
minimized



Inter-cluster
distances are
maximized



kita ga punya ground-truth untuk menyatakan klo label tersebut benar di titik tersebut

yg bener: jarak antar data yang di **asumsikan** berada di titik yang sama, kita ga penrah bener bener tau data tersebut ada di kelas yang mana

intra: jarak antar item

inter: jarak antar cluster

klo sesuatu sifatnya terukur maka itu harus ter-grounded

klo kita pake objek di inter clunter, objek ini sifatnya hipotetif → gak grounded → klo di ukur → menjadi hasilnya semu, maka ga bisa jarak antar cluster, pakelah jarak antar data

CLUSTER ANALYSIS: BASIC CONCEPTS (2)

- Clustering can be achieved by various clustering algorithms that differ significantly in their notion of what constitutes a cluster and how to efficiently find them.
- Popular notions of clusters include groups with small distances among the cluster members, dense areas of the data space, intervals, or particular statistical distributions.
- Typical applications
 - As a stand-alone tool to get insight into data distribution
 - As a preprocessing step for other algorithms
- Cluster analysis is unsupervised learning, in which data objects are unlabeled, no predefined classes.

APPLICATIONS OF CLUSTER ANALYSIS

- Data reduction
 - Summarization: Reduce the size of large data sets, preprocessing for regression, PCA, classification, and association analysis
 - Compression: Image processing: vector quantization
- Hypothesis generation and testing
- Prediction based on groups
 - Cluster & find characteristics/patterns for each group
- Finding K-nearest neighbors
 - Localizing search to one or a small number of clusters
- Outlier detection: Outliers are often viewed as those “far away” from any cluster

manfaat clustering:

- menemukan pola
- data reduction/dimentional reduction

- ... kardinalitas data, klo turun yang terdampak biaya komputasi
- recommendation
- untuk lokalisasi → untuk sampling → search space
-

RAG (Retrieval, Augmented, Generate)

flow AI: doc → R → vector → A → LLM → G → text

butuh clustering untuk mempercepat generate textnya

clustering menyelesaikan masalah kompleksitas komputasi (space)

WHAT IS NOT CLUSTER ANALYSIS?

- Supervised classification
 - Have class label information
- Simple segmentation
 - Dividing students into different registration groups alphabetically, by last name
- Results of a query
 - Groupings are a result of an external specification
- Graph partitioning
 - Some mutual relevance and synergy, but areas are not identical

QUALITY: WHAT IS GOOD CLUSTERING?

- A good clustering method will produce high quality clusters
 - high intra-class similarity: **cohesive** within clusters
 - low inter-class similarity: **distinctive** between clusters
- The quality of a clustering method depends on
 - the similarity measure used by the method
 - its implementation, and
 - its ability to discover some or all of the hidden patterns

dia ga punya ground-truth

intra-class high → cluster cohesive

inter-class low → cluternya distinctive

TYPES OF CLUSTERING

- **Hierarchical approach:**
 - Create a hierarchical decomposition of the set of data (or objects) using some criterion
 - Typical methods: Diana, Agnes, BIRCH, CAMELEON
- **Partitioning approach:**
 - A division data objects into non-overlapping subsets (clusters) such that each data object is in exactly one subset.
 - Evaluate cluster by some criterion, e.g., minimizing the sum of square errors
 - Typical methods: k-means, k-medoids, CLARANS
- **Density-based approach:**
 - Based on connectivity and density functions
 - Typical methods: DBSACN, OPTICS, DenClue
- **Grid-based approach:**
 - based on a multiple-level granularity structure
 - Typical methods: STING, WaveCluster, CLIQUE

ini kebalik, harusnya hirarki baru bisa dilakukan setelah partioning

hirarkikal benefitnya bisa start small, ga harus ke supersetnya, bisa ke subsetnya

density based

kondisi ketika data den banget, ada yang sparse banget, ini jadi problem untuk density based

gridbase: dipake untuk data yang punya multiple distribusi function

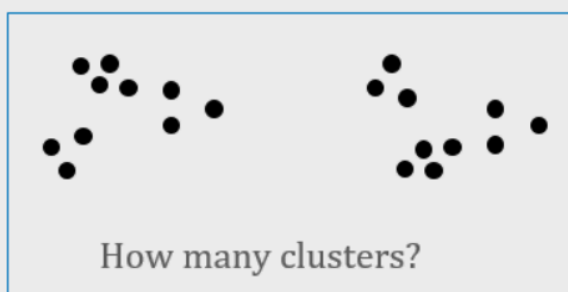
TYPES OF CLUSTERING (2)

- **Model-based:** A model is hypothesized for each of the clusters and tries to find the best fit of that model to each other
 - Fuzzy Clustering
 - Mixture Model
 - Self-Organizing Map (SOM)
- **Frequent pattern-based:** Based on the analysis of frequent patterns
 - Typical methods: p-Cluster
- **User-guided or constraint-based:** Clustering by considering user-specified or application-specific constraints
 - Typical methods: COD (obstacles), constrained clustering
- **Link-based clustering:**
 - Objects are often linked together in various ways
 - Massive links can be used to cluster objects: SimRank, LinkClus

mixture bisa macem macem, intinya menggabungkan beberapa model untuk bisa capture

SOM pakai neural network

NOTION OF A CLUSTER CAN BE AMBIGUOUS



semakin besar clusternya → karakteristiknya semakin detail → cost besar

semakin kecil jumlah cluster → lebih umum → cost rendah

CHARACTERISTICS OF THE INPUT DATA ARE IMPORTANT

- Dimensionality
- Noise and Outliers
- Type of Distribution
- Type of proximity or density measure
 - This is a derived measure, but central to clustering
- Sparseness
 - Dictates type of similarity
 - Adds to efficiency
- Attribute type
 - Dictates type of similarity
- Type of Data
 - Dictates type of similarity
 - Other characteristics, e.g., autocorrelation

clustering algonya itu sangat sensitif, maka perhatikan:

- dimensionality: semakin besar → dimensi semakin besar, hitung untung ruginya, klo besar di reduksi bisa baris atau kolom
- noise dan outlier: karena centroid based on rata"
- tipe distribusi: contoh di dbSCAN klo skew nya besar/SD besar → perlu diperhatikan apakah algonya cocok
- proximity/density measure: melihat ketidak miripan sebuah item

- sparseness

PROXIMITY MEASURE

- Refers to a data similarity and dissimilarity measure
- **Similarity**
 - Numerical measure of how alike two data objects are
 - Value is higher when objects are more alike
 - Often falls in the range $[0,1]$
- **Dissimilarity** (e.g., distance measure)
 - Numerical measure of how different two data objects are
 - Lower when objects are more alike
 - Minimum dissimilarity is often 0
 - Upper limit varies

clustering itu berkaitan dengan cohesive dan distinctness

untuk mengukur itu semua kita butuh matrix

misal cari similarity: cari yang mirip yang mereka miliki sama

dissimilarity: tinggal cari yang beda

distance itu sebenarnya bukan nyari yang mirip, karena ga ada absolute zero nya, tapi dissimilarity, makanya klo dia dengan dirinya sendiri akan menghasilkan distance = 0

PROXIMITY MEASURE (2)

- **Data matrix**

- Used to store the data objects
- n data points with p dimensions
- Two modes

- **Dissimilarity matrix**

- Used to store dissimilarity values for pairs of objects
- n data points, but registers only the distance
- A triangular matrix
- Single mode

- Object dissimilarity **computation** based on the **attribute type** of objects

$$\begin{bmatrix} x_{11} & \dots & x_{1f} & \dots & x_{1p} \\ \dots & \dots & \dots & \dots & \dots \\ x_{i1} & \dots & x_{if} & \dots & x_{ip} \\ \dots & \dots & \dots & \dots & \dots \\ x_{n1} & \dots & x_{nf} & \dots & x_{np} \end{bmatrix}$$

$$\begin{bmatrix} 0 & & & & \\ d(2,1) & 0 & & & \\ d(3,1) & d(3,2) & 0 & & \\ \vdots & \vdots & \vdots & \ddots & \\ d(n,1) & d(n,2) & \dots & \dots & 0 \end{bmatrix}$$

untuk mengukur dissimilarity pake proximity measure

PROXIMITY MEASURE (3)

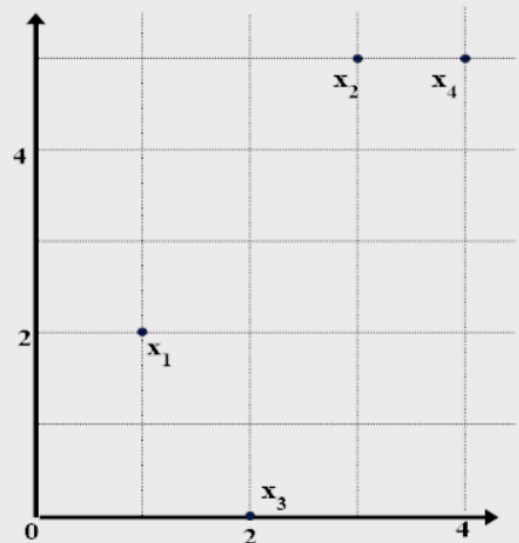
- Example:

- Data matrix

Point	Attr. 1	Attr. 2
x_1	1	2
x_2	3	5
x_3	2	0
x_4	4	5

- Dissimilarity matrix (with Euclidean distance)

	x_1	x_2	x_3	x_4
x_1	0			
x_2	3.61	0		
x_3	2.24	5.1	0	
x_4	4.24	1	5.39	0



PROXIMITY MEASURE FOR NUMERIC ATTRIBUTES

- **Minkowski distance:**

$$d(i, j) = \sqrt[h]{|x_{i1} - x_{j1}|^h + |x_{i2} - x_{j2}|^h + \cdots + |x_{ip} - x_{jp}|^h}$$

where $i = (x_{i1}, x_{i2}, \dots, x_{ip})$ and $j = (x_{j1}, x_{j2}, \dots, x_{jp})$ are two p -dimensional data objects, and h is the order (the distance so defined is also called $L - h$ norm)

- Properties
 - $d(i, j) > 0$ if $i \neq j$, and $d(i, j) = 0$ (*positive definiteness*)
 - $d(i, j) = d(j, i)$ (*symmetry*)
 - $d(i, j) \leq d(i, k) + d(k, j)$ (*triangle inequality*)
- A distance that satisfies these properties is a **metric**

PROXIMITY MEASURE FOR NUMERIC ATTRIBUTES (2)

- Special case of **Minkowski** distance:
- $h = 1$, **Manhattan** (city block, L_1 norm) distance

$$d(i, j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \cdots + |x_{ip} - x_{jp}|$$

- $h = 2$, **Euclidean** distance (L_2 norm)

$$d(i, j) = \sqrt{|x_{i1} - x_{j1}|^2 + |x_{i2} - x_{j2}|^2 + \cdots + |x_{ip} - x_{jp}|^2}$$

- $h = \infty$, **supremum** distance (L_{max} norm, L_∞ norm)

$$d(i, j) = \lim_{h \rightarrow \infty} \left(\sum_{f=1}^p |x_{if} - x_{jf}|^h \right)^{\frac{1}{h}} = \max_f |x_{if} - x_{jf}|$$

PROXIMITY MEASURE FOR NUMERIC ATTRIBUTES (3)

- Example:

Point	Attr. 1	Attr. 2
x_1	1	2
x_2	3	5
x_3	2	0
x_4	4	5

Data matrix

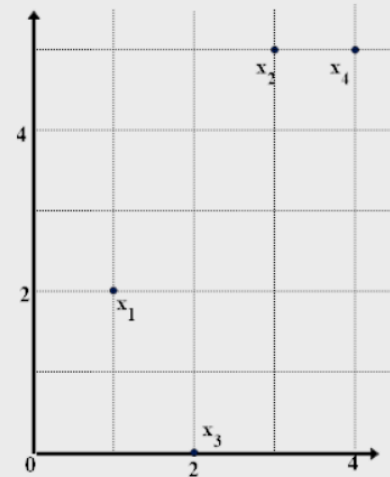
L_2	x_1	x_2	x_3	x_4
x_1	0			
x_2	3.61	0		
x_3	2.24	5.1	0	
x_4	4.24	1	5.39	0

Dissimilarity matrix
(with Euclidean)

Dissimilarity matrix
(with Manhattan)

L	x_1	x_2	x_3	x_4
x_1	0			
x_2	5	0		
x_3	3	6	0	
x_4	6	1	7	0

Dissimilarity matrix
(with supremum)



lebih mudah manhattan → karena komputer lebih suka tambah kurang daripada kali bagi

PROXIMITY MEASURE FOR ORDINAL ATTRIBUTES

- An ordinal attribute can be discrete or continuous
- Order is important, e.g., rank
- Similar to numeric attribute that can be treated like interval-scaled:
 - Replace x_{if} by their rank $r_{if} \in \{1, \dots, M_f\}$
 - Map the range of each attribute onto $[0, 1]$ by replacing i -th object in the f -th attribute by

$$z_{if} = \frac{r_{if} - 1}{M_f - 1}$$
 - Compute the dissimilarity using methods for interval-scaled attributes

proximity measure juga bisa dipakai untuk ordinal ... bukan yang numerical
pastikan pake skala yang tepat

PROXIMITY MEASURE FOR MIXED ATTRIBUTES TYPE

- A database may contain all attribute types
 - Nominal, symmetric binary, asymmetric binary, numeric, ordinal
- One may use a weighted formula to combine their effects

$$d(i, j) = \frac{\sum_{f=1}^p \delta_{ij}^{(f)} d_{ij}^{(f)}}{\sum_{f=1}^p \delta_{ij}^{(f)}}$$

- f is binary or nominal: $d_{ij}^{(f)} = 0$ if $x_{if} = x_{jf}$, or $d_{ij}^{(f)} = 1$ otherwise
- f is numeric: use the normalized distance
- f is ordinal:
 - Compute ranks r_{if} and $z_{if} = \frac{r_{if}-1}{M_f-1}$
 - Treat z_{if} as interval-scaled

PARTITIONING ALGORITHMS

- Partitioning method: partitioning a database D of n objects into a set of k clusters, such that the sum of squared distances is minimized (where c_i is the centroid or medoid of cluster C_i)

$$E(D, \mu) = \sum_{j=1}^k \sum_{i=1}^N d(x_i, \mu_j)^2$$

- The clustering find a partition of k clusters that optimizes the certain partitioning criterion. Methods:
 - **k-Means** (MacQueen' 67, Lloyd' 57/'82): each cluster is represented by center of cluster
 - **k-Medoids** or PAM (Kaufmann & Rousseeuw' 87): each cluster is represented by one of the objects in the cluster
 - Exhaustively enumerate all partitions

kmeans problemnya dia sensitif terhadap outlier sehingga muncul kmedoids

k-MEANS CLUSTERING

- Creates an initial set of k partitions
- It then uses an iterative relocation technique that attempts to improve the partitioning by moving objects from one group to another.
- Each cluster is associated with a **centroid** (center point)
- Clustering: each point is assigned to the cluster with the closest centroid

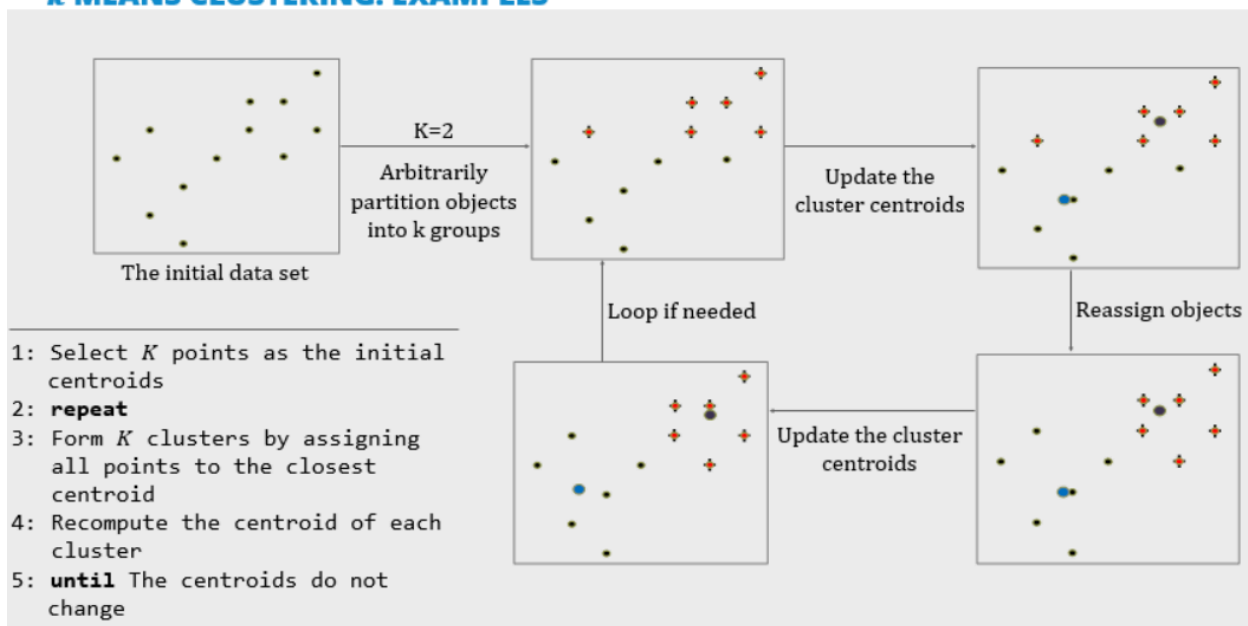
```
1: Select  $K$  points as the initial centroids
2: repeat
3:     Form  $K$  clusters by assigning all points to the closest centroid
4:     Recompute the centroid of each cluster
5: until The centroids do not change
```

intitinya di hitung → di clustering → ulangi sampai hasil clusternhya ga berubah ubah lagi

k-MEANS CLUSTERING (2)

- Initial centroids are often chosen randomly
 - Clusters produced vary from one run to another
- The centroid is (typically) the mean of the points in the cluster
- 'Closeness' is measured by **Euclidean distance, cosine similarity, correlation**, etc.
- *k*-means will converge for common similarity measures mentioned above
- Most of the convergence happens in the first few iterations
 - Often the stopping condition is changed to 'Until relatively few points change clusters'
- Complexity is $O(n*K*I*d)$
 - n = number of points, K = number of clusters,
 - I = number of iterations, d = number of attributes

k-MEANS CLUSTERING: EXAMPLES



COMMENTS ON THE *k*-MEANS METHOD

- **Strength:** Efficient: $O(tkn)$, where n is # objects, k is # clusters, and t is # iterations. Normally, $k, t \ll n$.
 - Comparing: PAM: $O(k(n-k)^2)$, CLARA: $O(k^2 + k(n-k))$
- **Comment:** Often terminates at a local optimal
- **Weakness:**
 - Applicable only to objects in a continuous n -dimensional space
 - Using the k -modes method for categorical data
 - In comparison, k -medoids can be applied to a wide range of data
 - Need to specify k , the number of clusters, in advance (there are ways to automatically determine the best k (see Hastie et al., 2009))
 - Sensitive to noisy data and outliers
 - Since an object with an extremely large value may substantially distort the distribution of the data
 - Not suitable to discover clusters with non-convex shapes

minusnya juga sering terjebak local optima (solusi terbaik di lokasi tersbeut), padahal yang di cari global optima

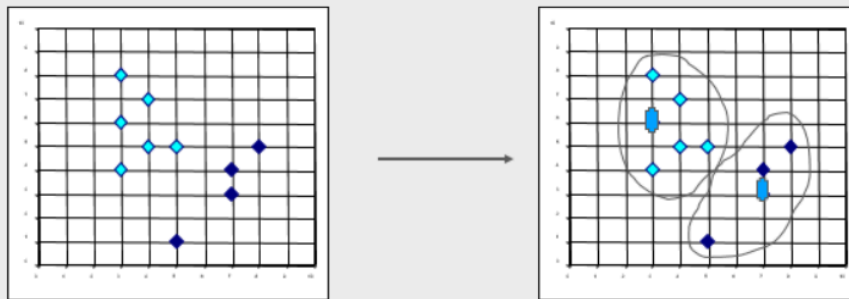
VARIATIONS OF THE *k*-MEANS METHOD

- Most of the variants of the k -means which differ in
 - Selection of the initial k means
 - Dissimilarity calculations
 - Strategies to calculate cluster means
- Handling categorical data: k -modes
 - Replacing means of clusters with modes
 - Using new dissimilarity measures to deal with categorical objects
 - Using a frequency-based method to update modes of clusters
 - A mixture of categorical and numerical data: k -prototype method

klo data kategorikal → pake modus

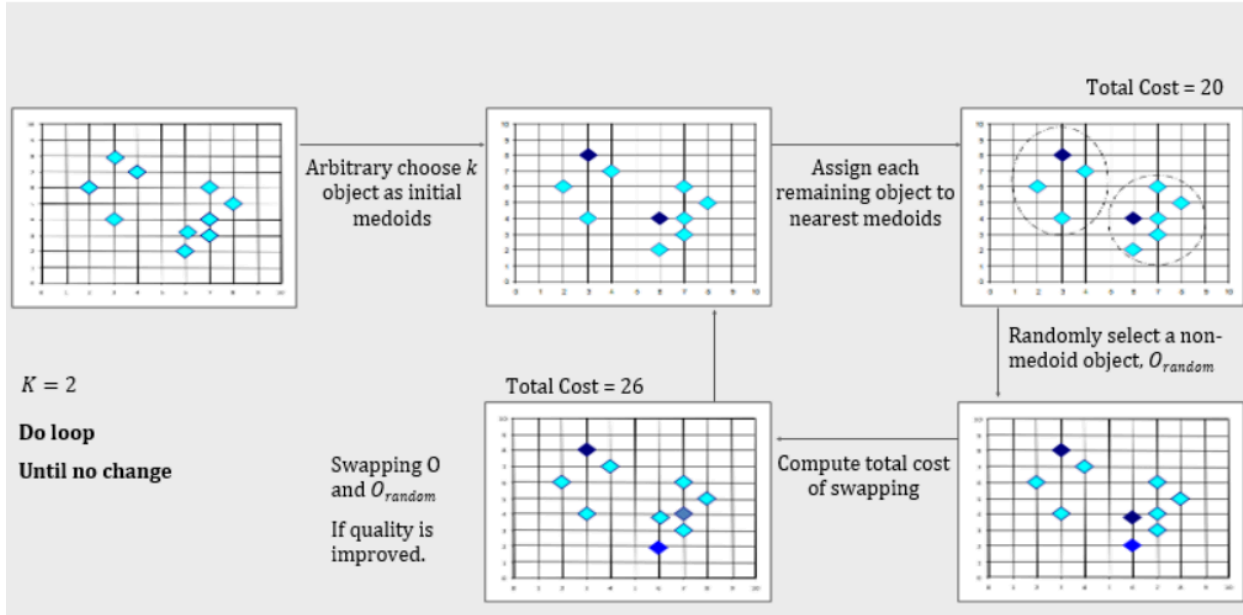
k-MEDOIDS

- Instead of taking the mean value of the object in a cluster as a reference point, **medoids** can be used, which is the **most centrally located** object in a cluster



daripada pakai titik centroid imajiner, dia pakai data point sebagai representatif di situ

k-MEDOIDS CLUSTERING ALGORITHM



k-MEDOIDS CLUSTERING ALGORITHM (2)

- *K-Medoids* Clustering: Find representative objects (**medoids**) in clusters
 - *PAM* (Partitioning Around Medoids, Kaufmann & Rousseeuw 1987)
 - Starts from an initial set of medoids, and iteratively replaces one of the medoids by one of the non-medoids if it improves the total distance of the resulting clustering
 - *PAM* works effectively for small datasets, but does not scale well for large datasets (due to the computational complexity)
 - Efficiency improvement on PAM
 - *CLARA* (Kaufmann & Rousseeuw, 1990): PAM on samples
 - *CLARANS* (Ng & Han, 1994): Randomized re-sampling

pam bagus untuk small datasets, maka muncullah:

- CLARA: PAM on samples
- CLARANS: randomized re-sampling



lah sampelnya gimana taunya? kan clustering untuk sampling?

SUMMARY

- **Cluster analysis** groups objects based on their **similarity** and has wide applications
- Measure of similarity can be computed for **various types of data**
- Clustering algorithms can be **categorized** into partitioning methods, hierarchical methods, density-based methods, grid-based methods, and model-based methods
- **K-means** and **K-medoids** algorithms are popular partitioning-based clustering algorithms

EXERCISE

1. What is clustering in data mining? Explain with a simple real-life example.
2. Mention two common applications of clustering in real-world scenarios.
3. Suppose you are using K-Means clustering with $K = 3$. What will happen if the initial centroids are poorly chosen?